

CS & IT ENGINEERING



THEORY OF COMPUTATION

✓ REGULAR EXPRESSION

Lecture No.- 03



By- Venkat sir



Recap of Previous Lecture



Topic

Regular Expression Construction
?????

Properties of Reg Expr.



Topics to be Covered



Topic

Conversion from ϵ -NFA to NFA

Topic

??

$F.A \iff \text{Regular Expression}$

Topic

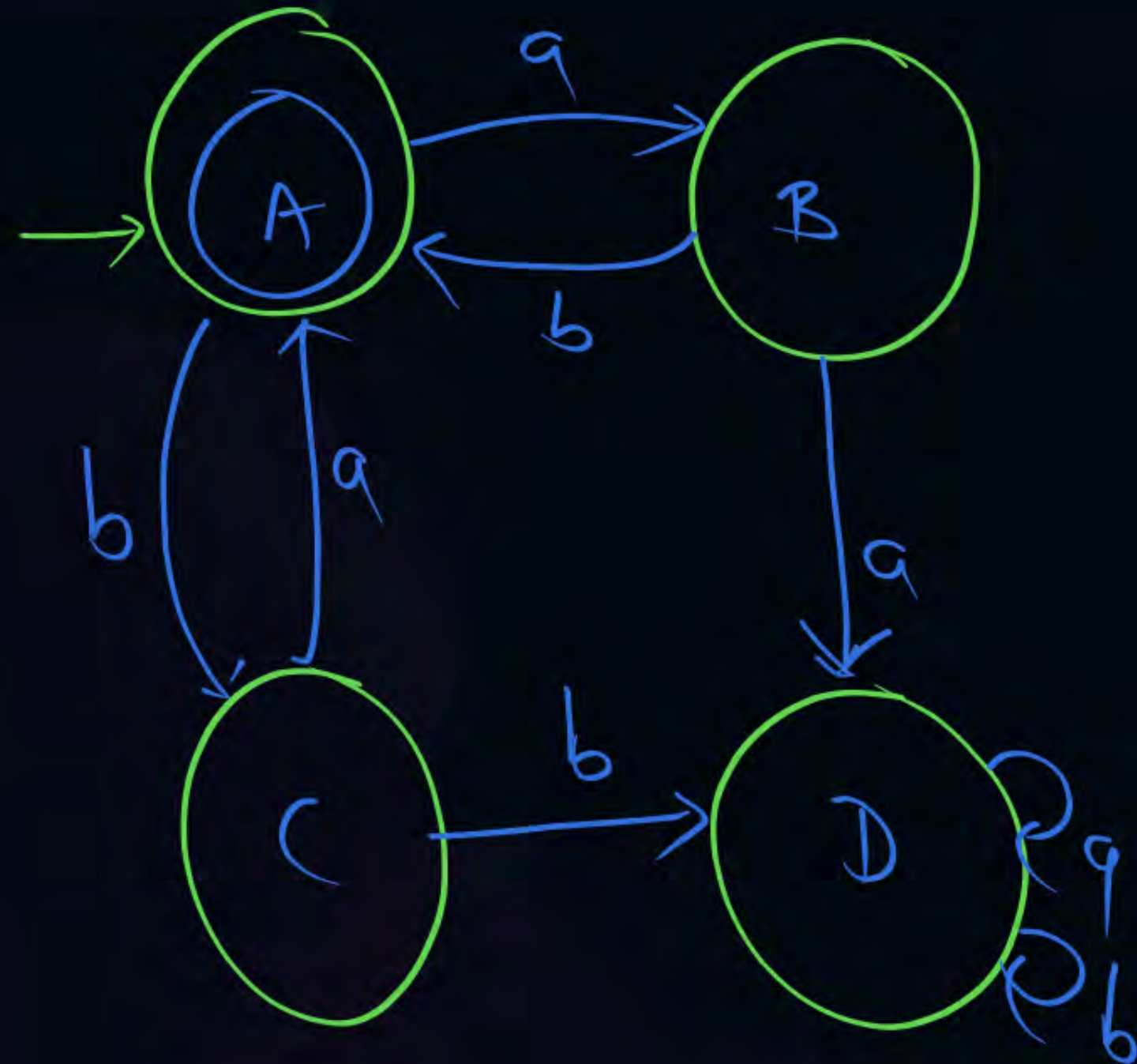
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Topic

??

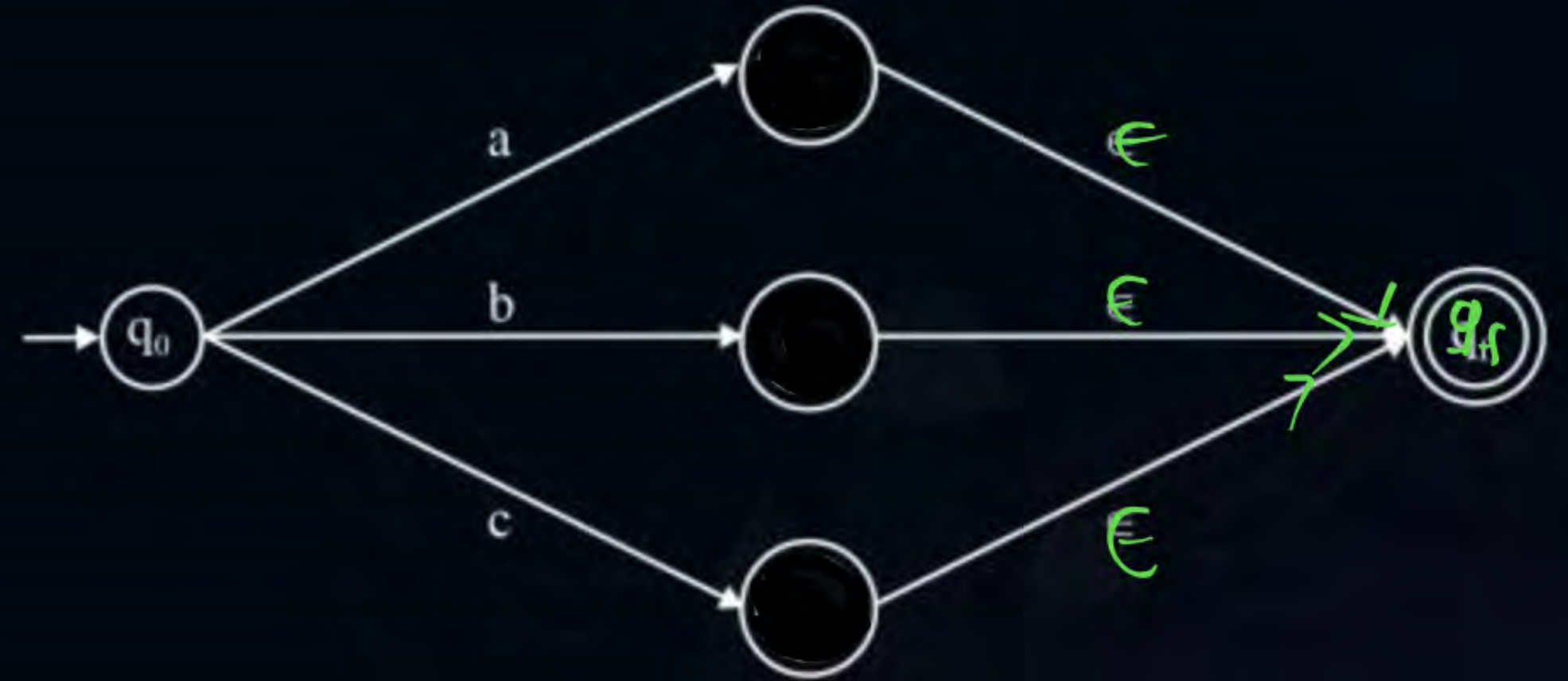
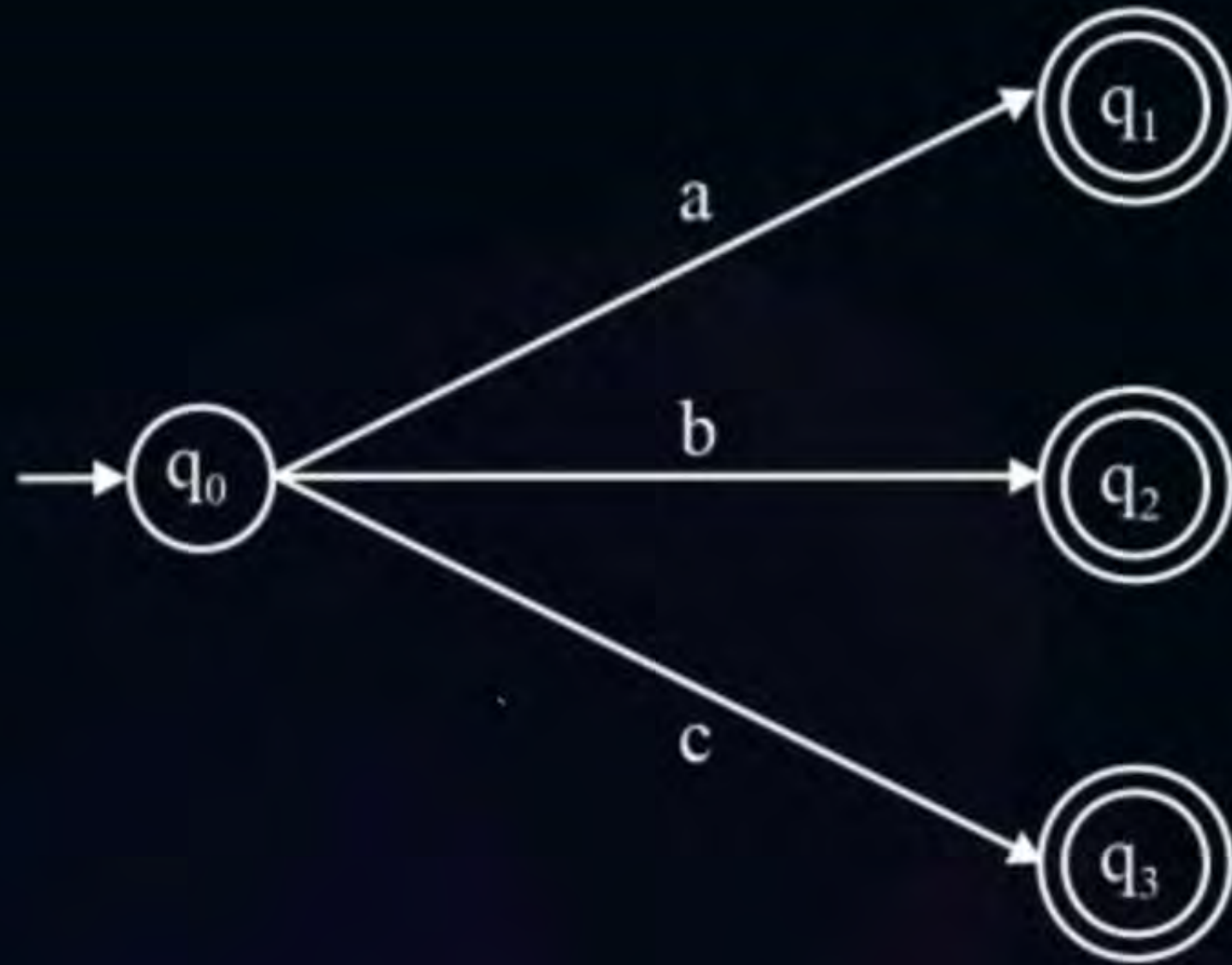
FINITE AUTOMATA TO REGULAR EXPRESSION

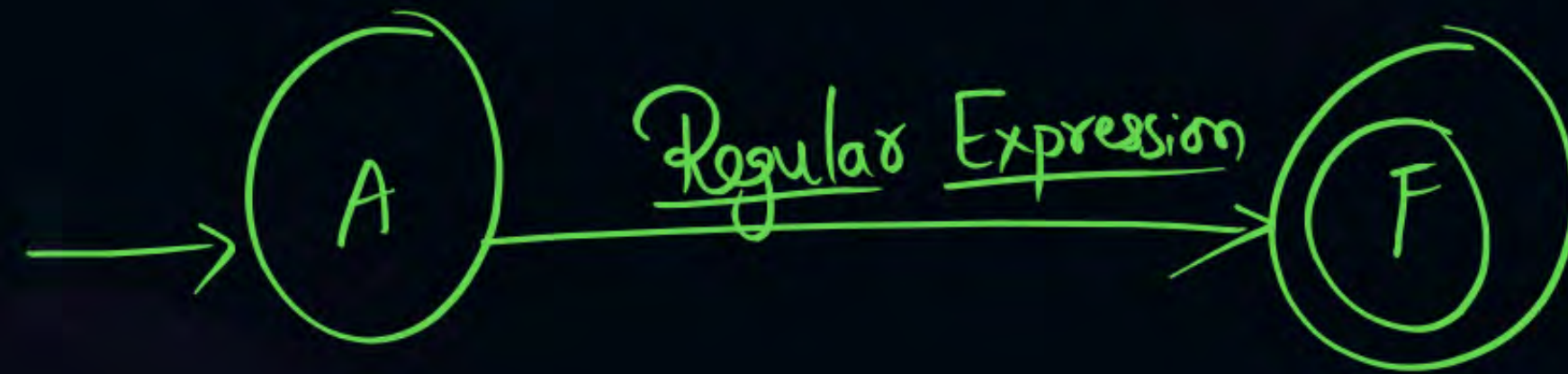
- ① state elimination method
- ② Arden's method



State Elimination method

1.

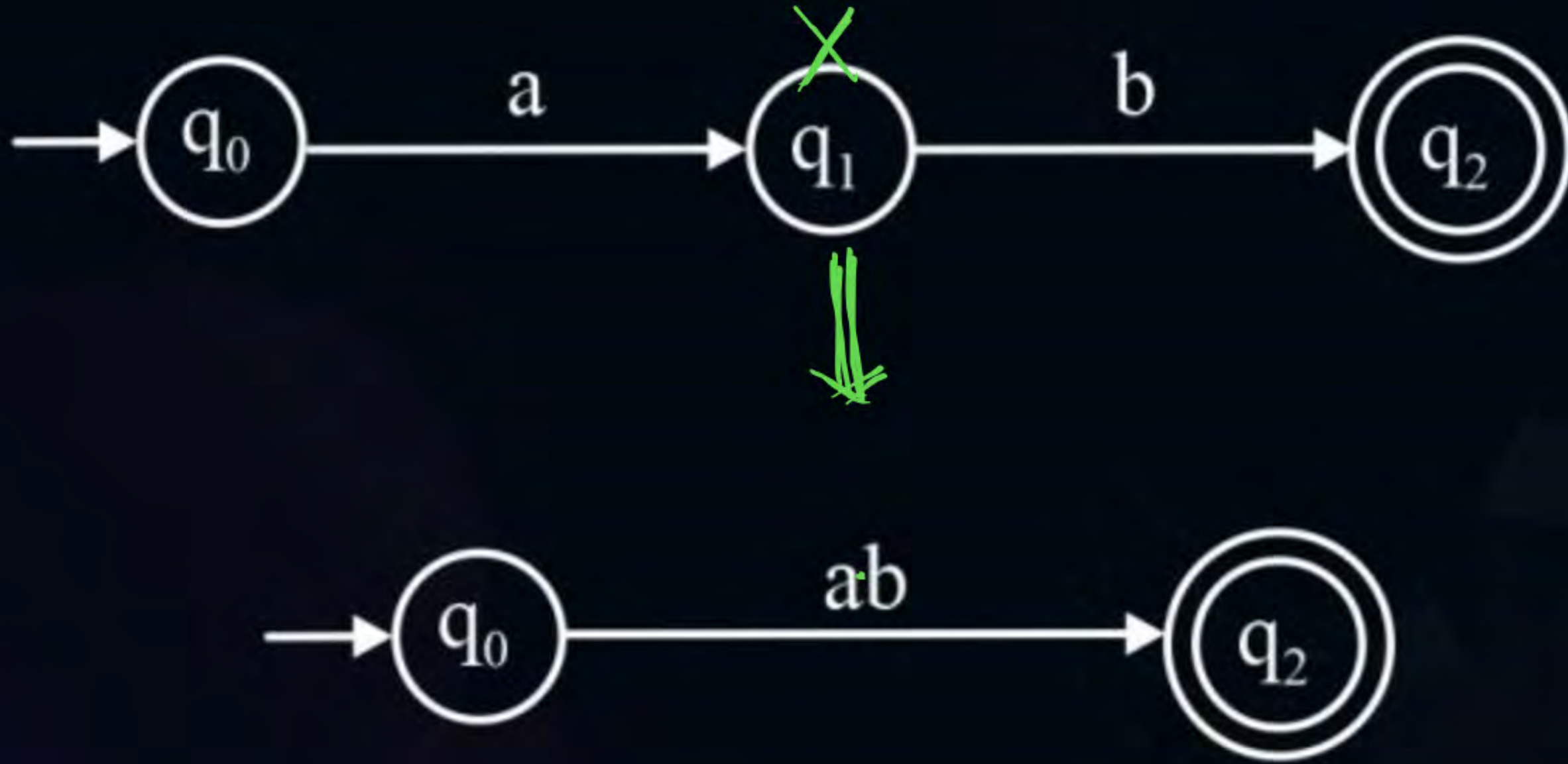




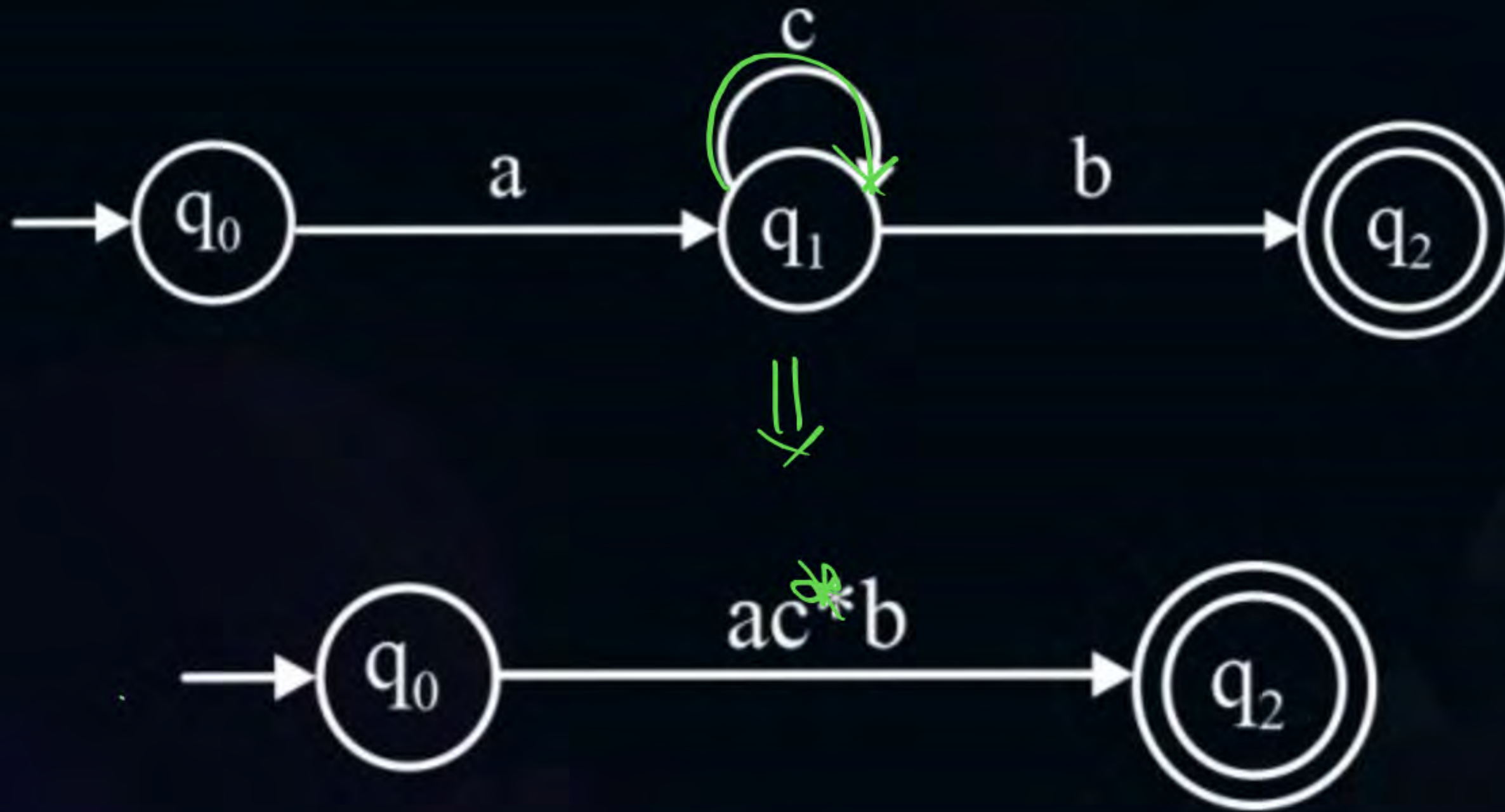
2



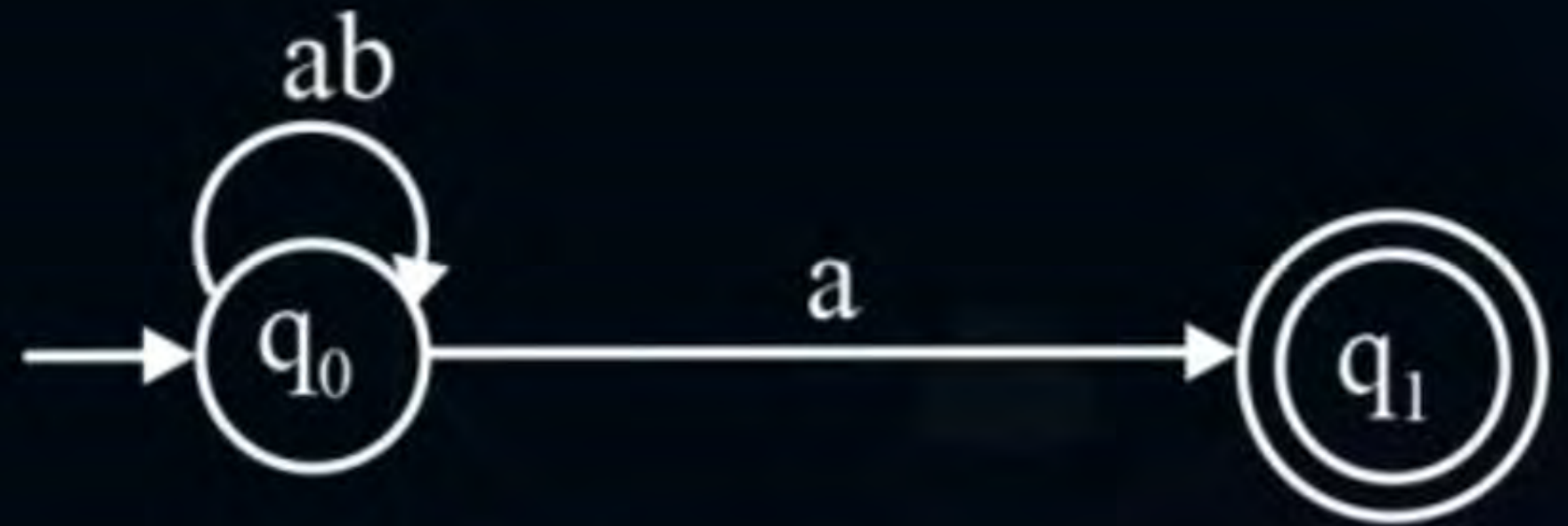
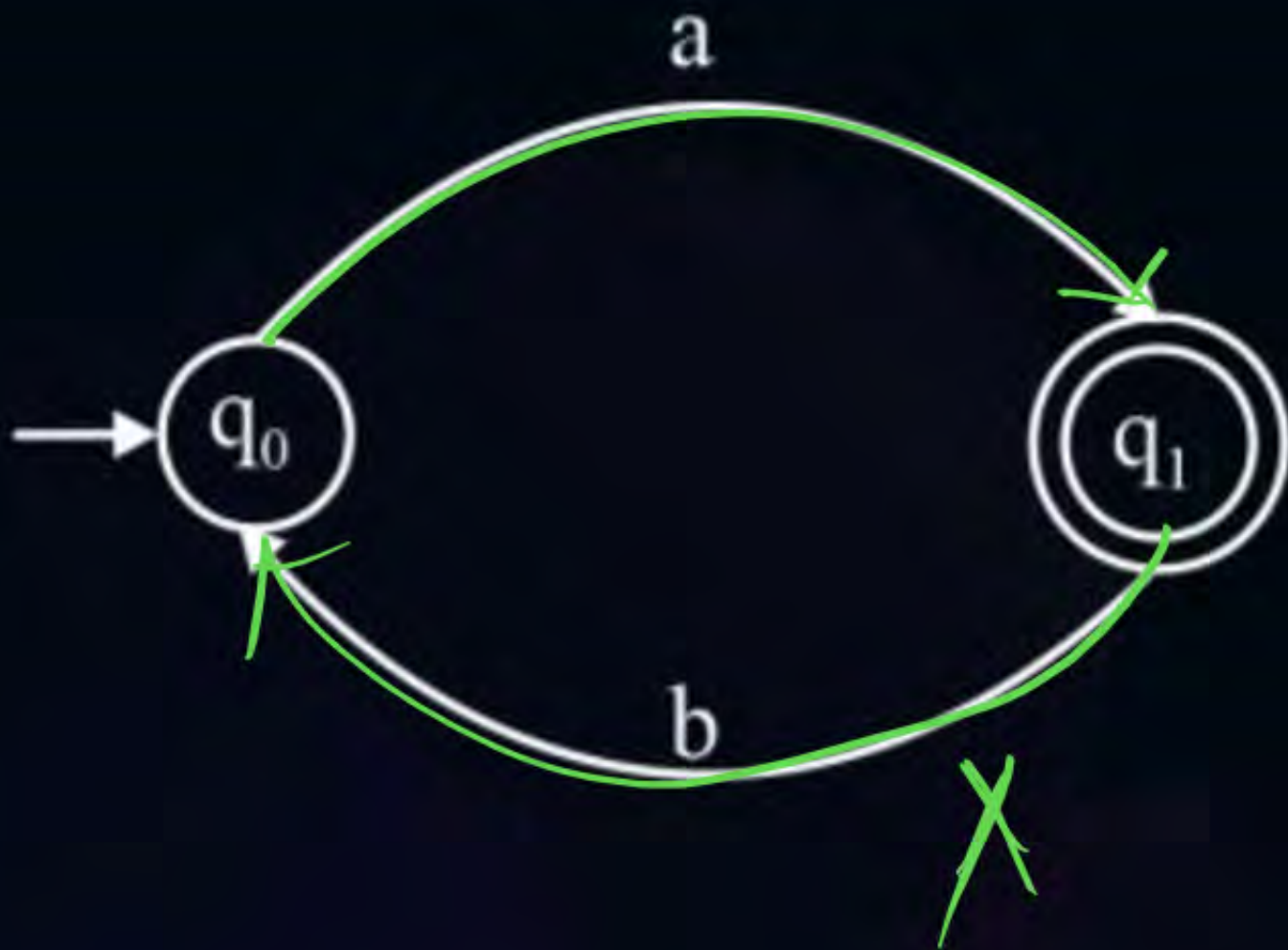
3.



4.



5.

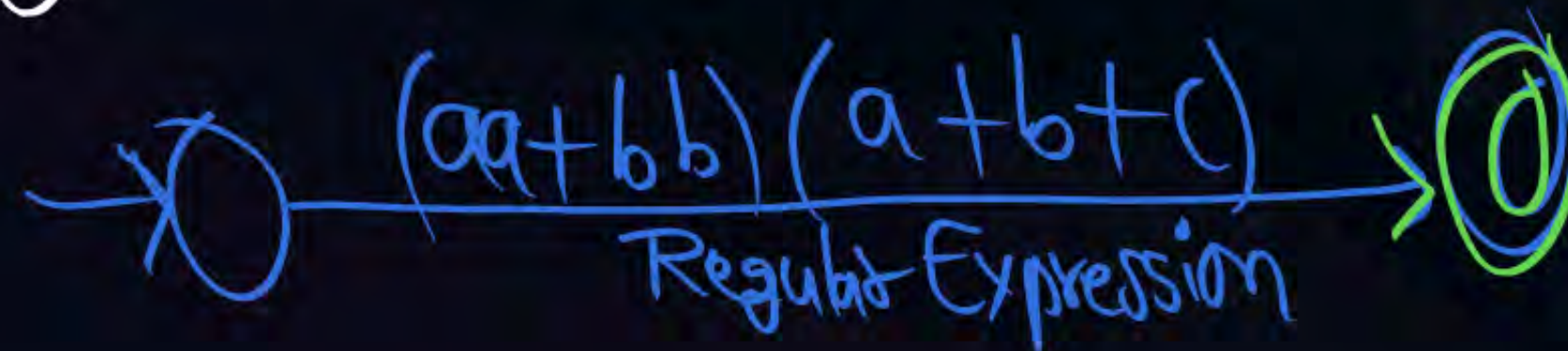
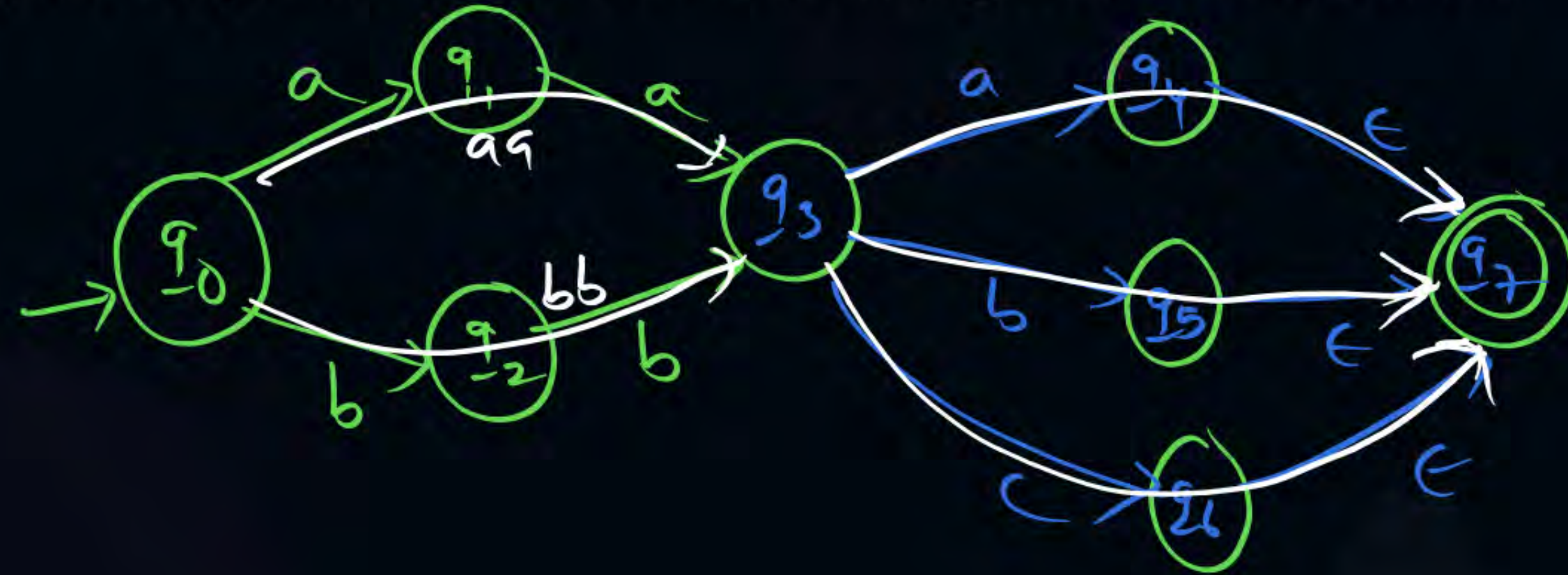


$(ab)^*$ $\cap$

#Q.

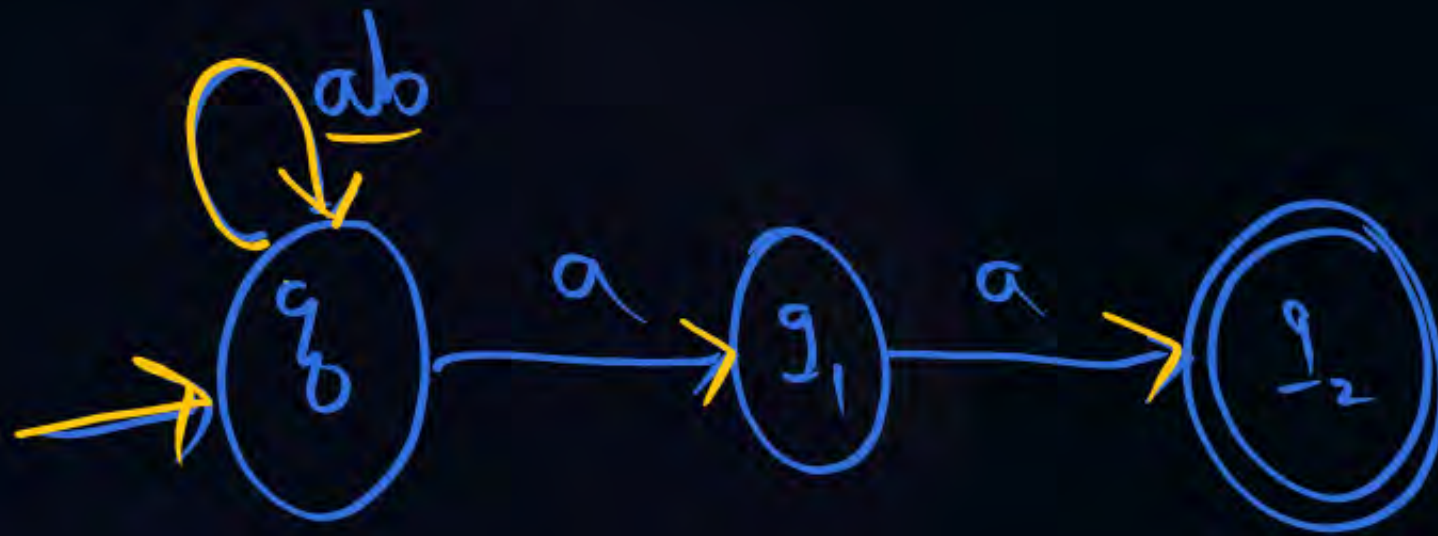
①

Construct Regular Expression for the following Finite Automata.



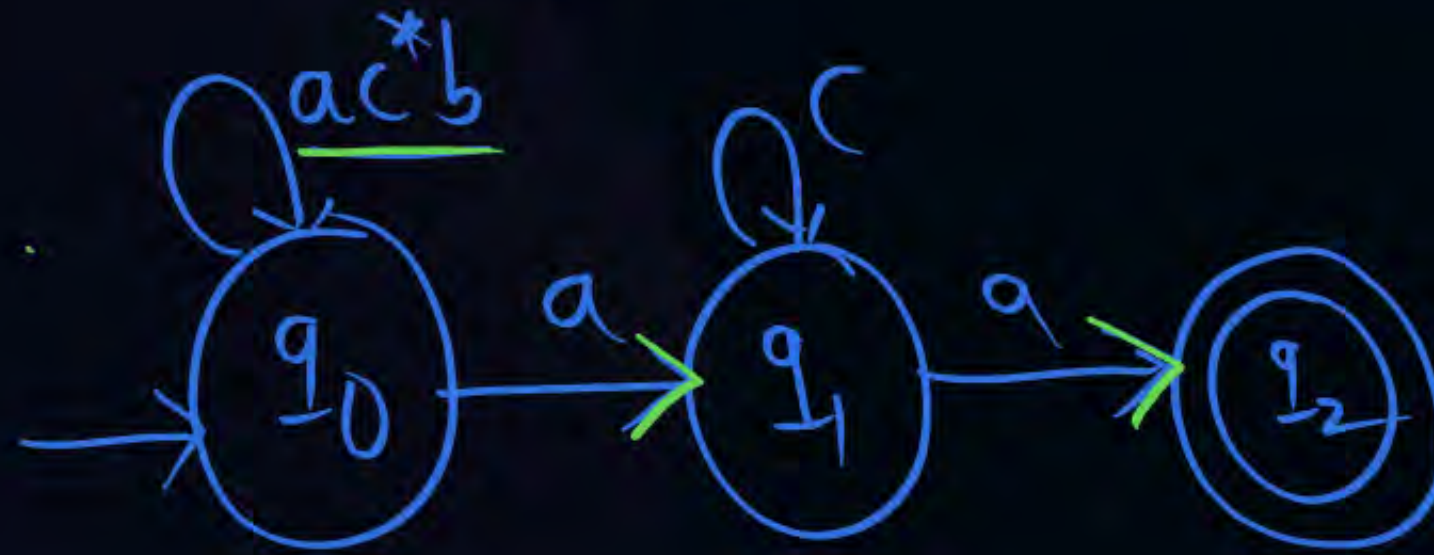
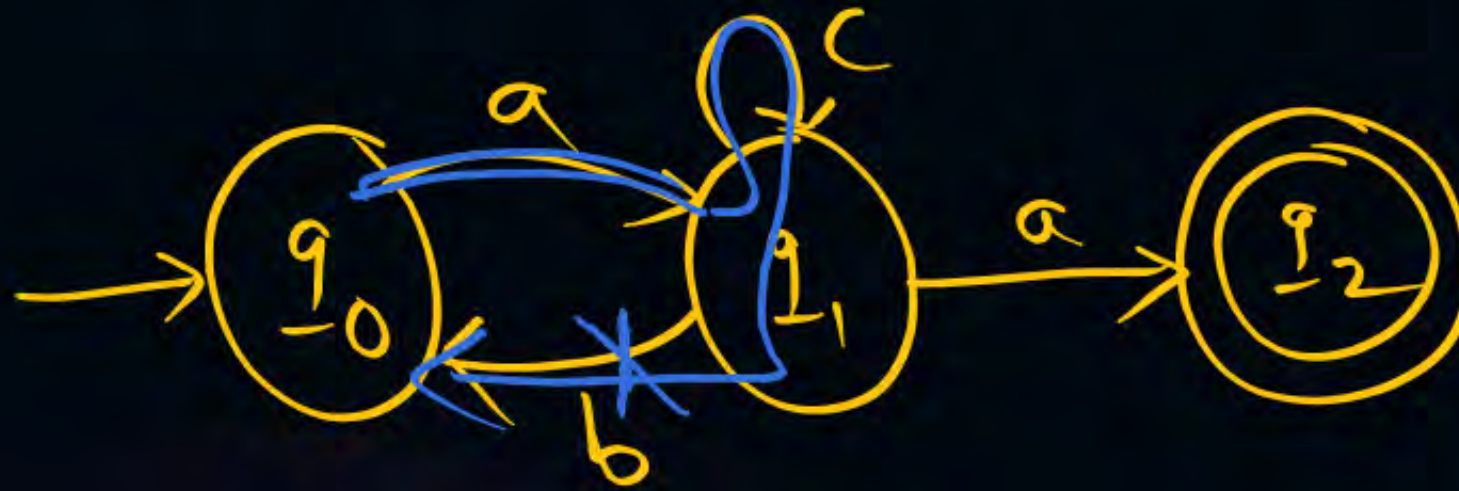
#Q. Construct Regular Expression for the following Finite Automata.

(2)



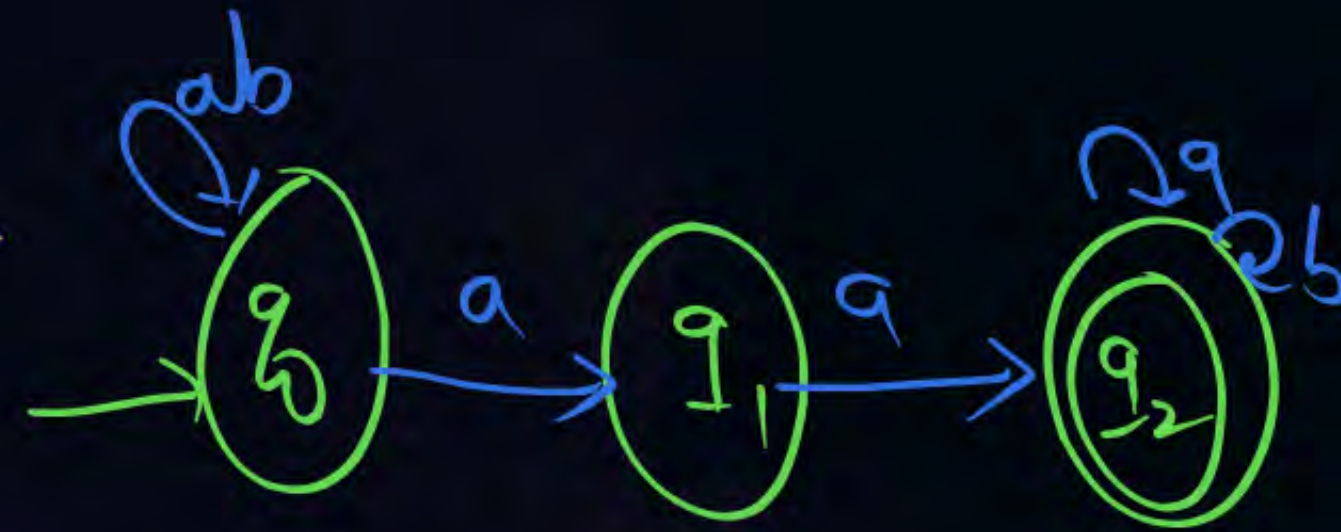
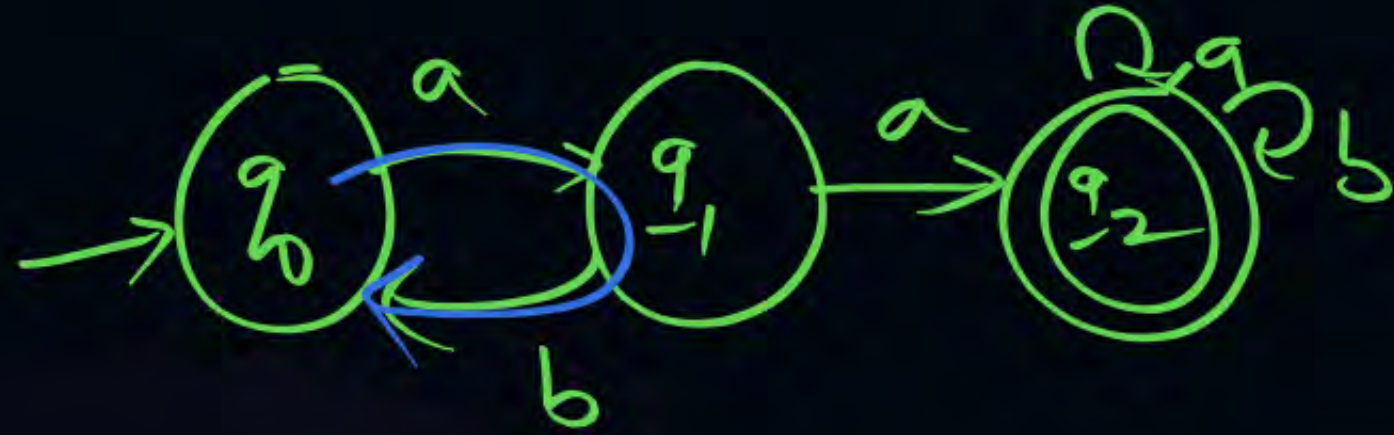
$(ab)^* aa$

#Q. Construct Regular Expression for the following Finite Automata.



$$\left\{ \underline{(ac^*b)^*} \underline{ac^*a} \right\}$$

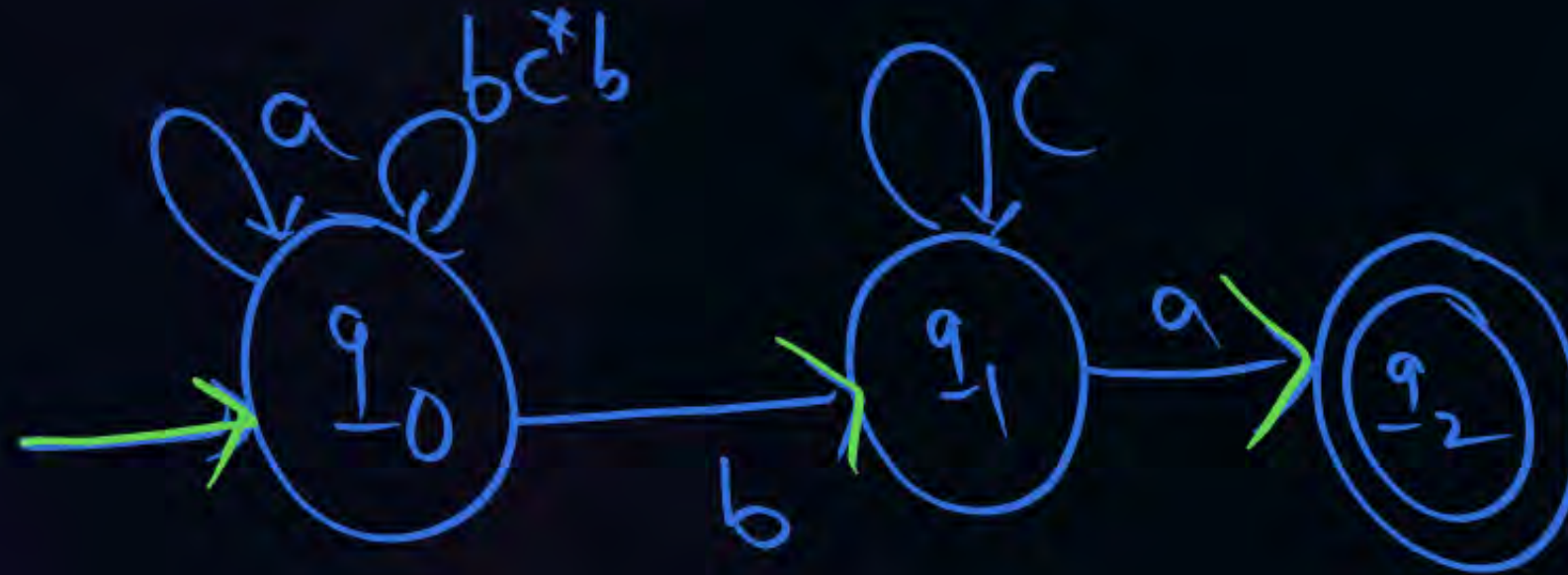
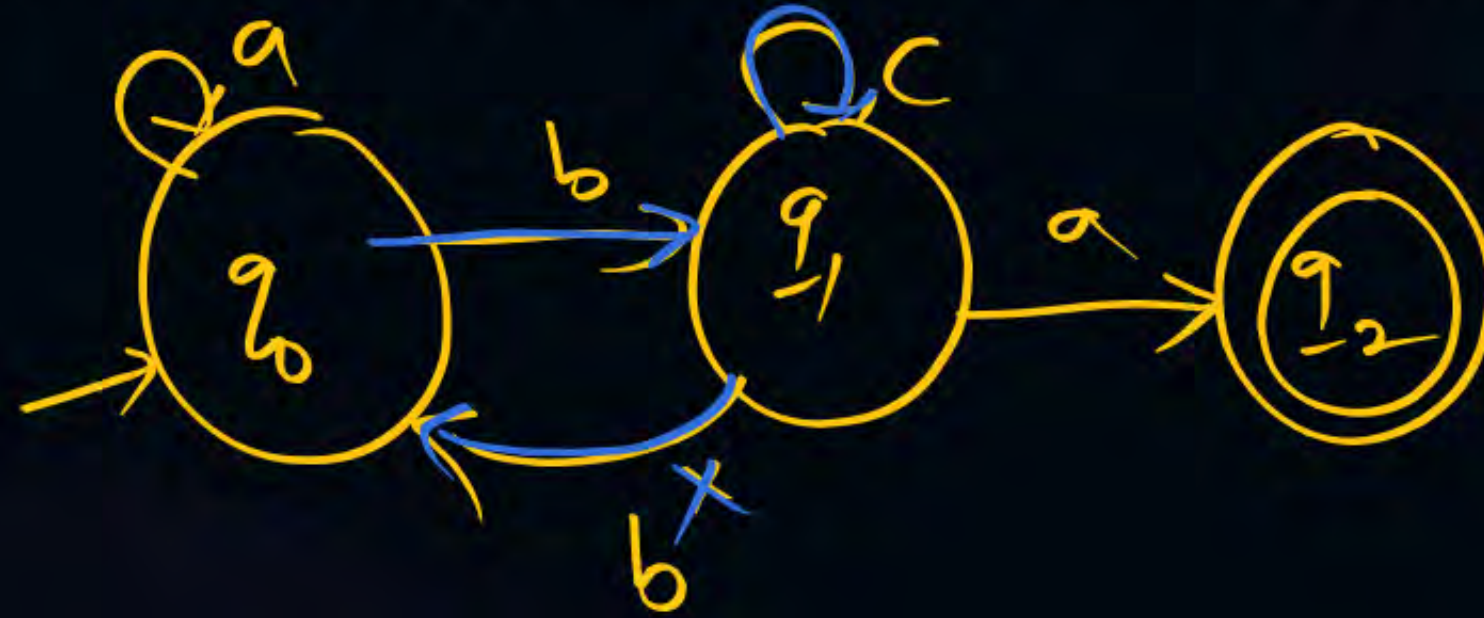
#Q. Construct Regular Expression for the following Finite Automata.



$$\underline{\underline{(ab)^* a a (a+b)^*}}$$



#Q. Construct Regular Expression for the following Finite Automata.



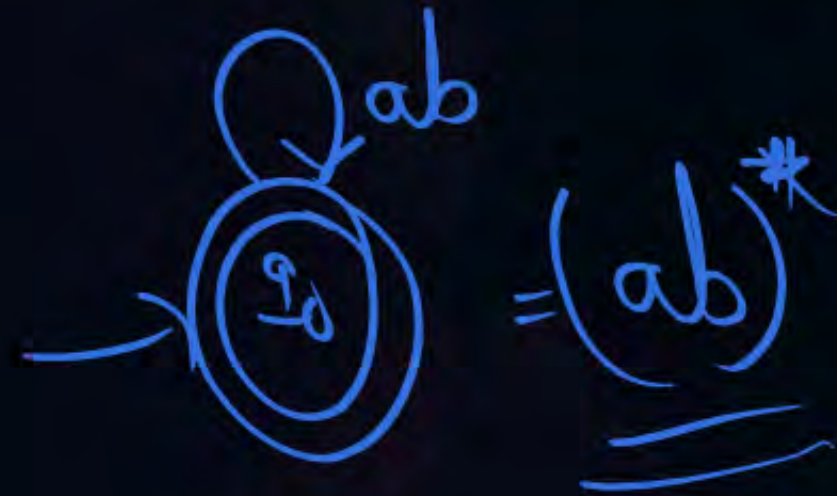
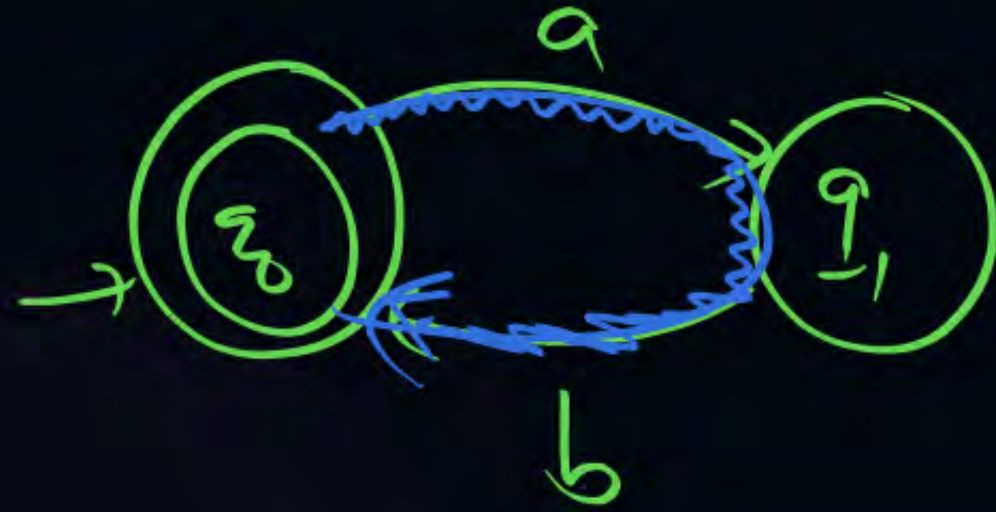
$$\underline{\underline{(a + bc^*b)^* bc^*a}}$$

[NAT]

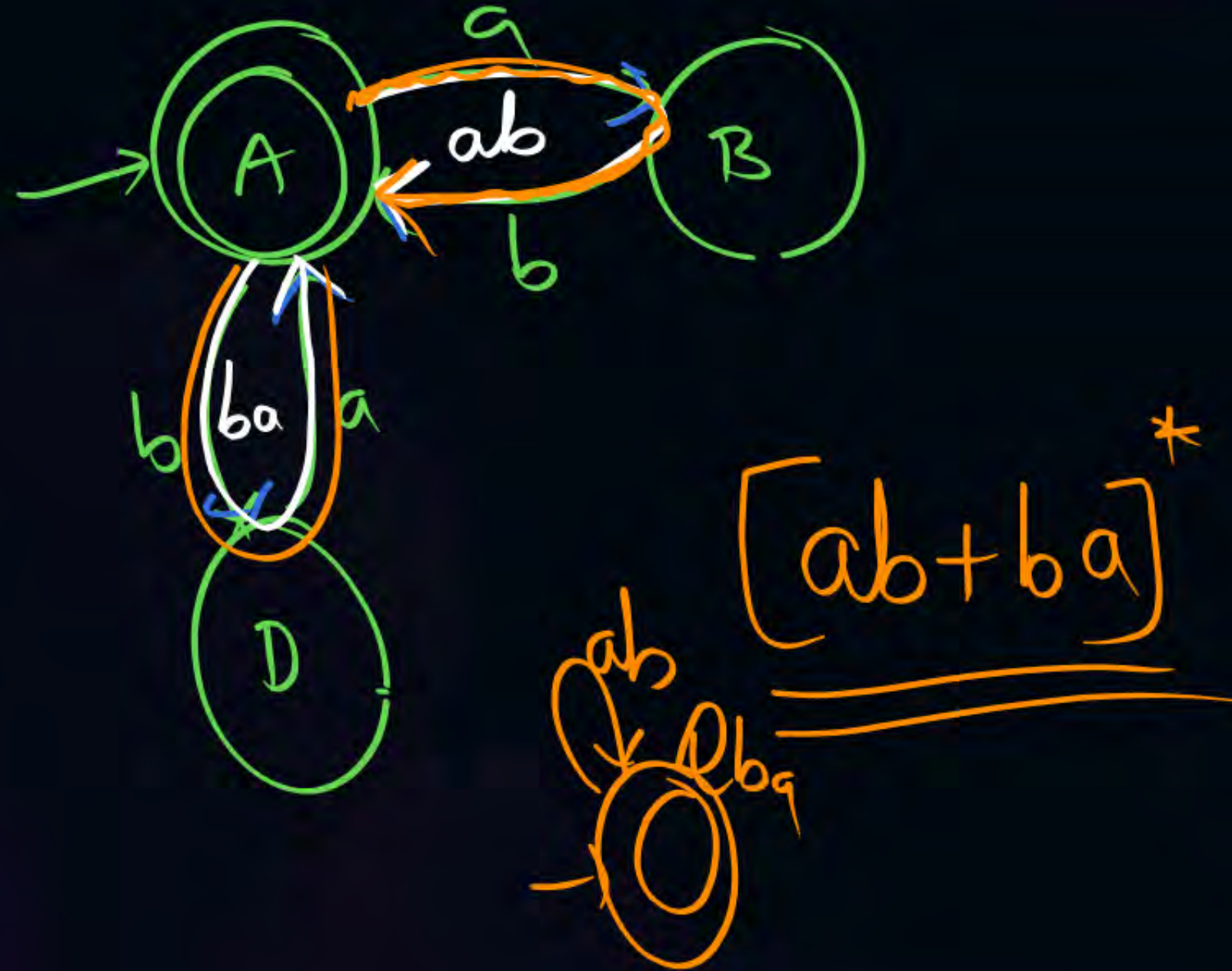
#Q. Construct Regular Expression for the following Finite Automata.



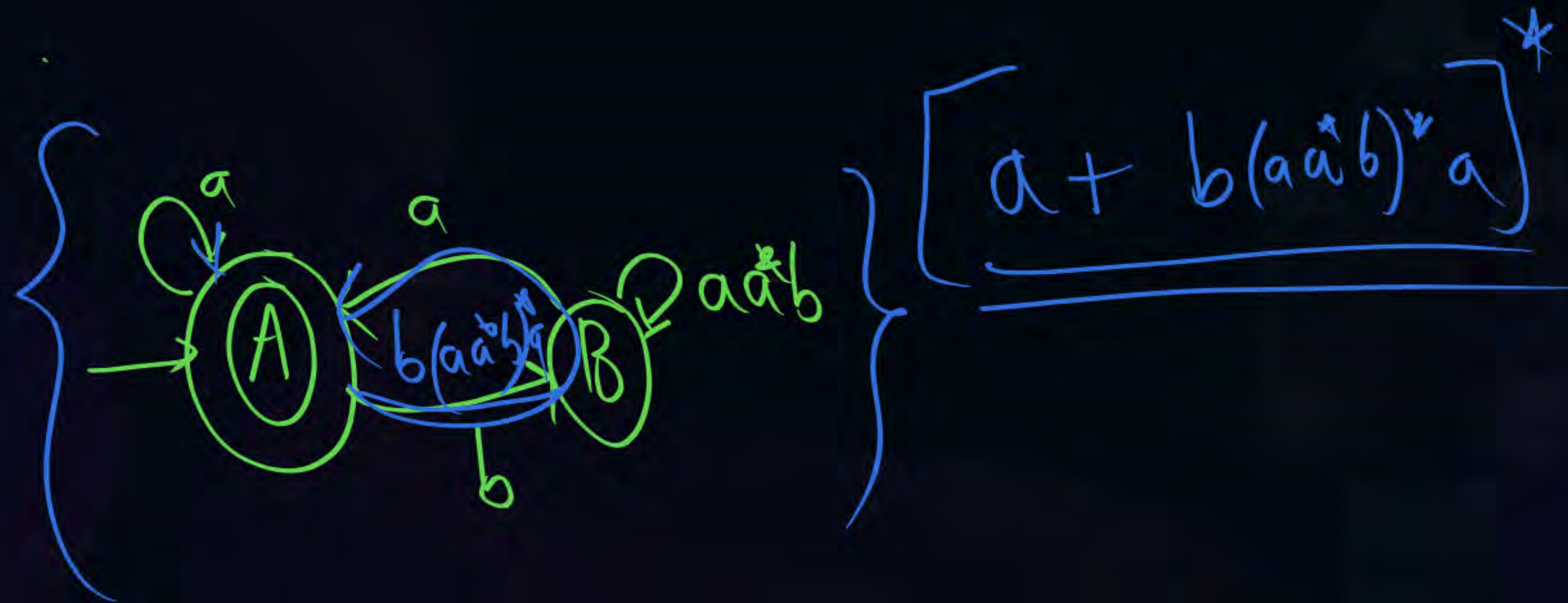
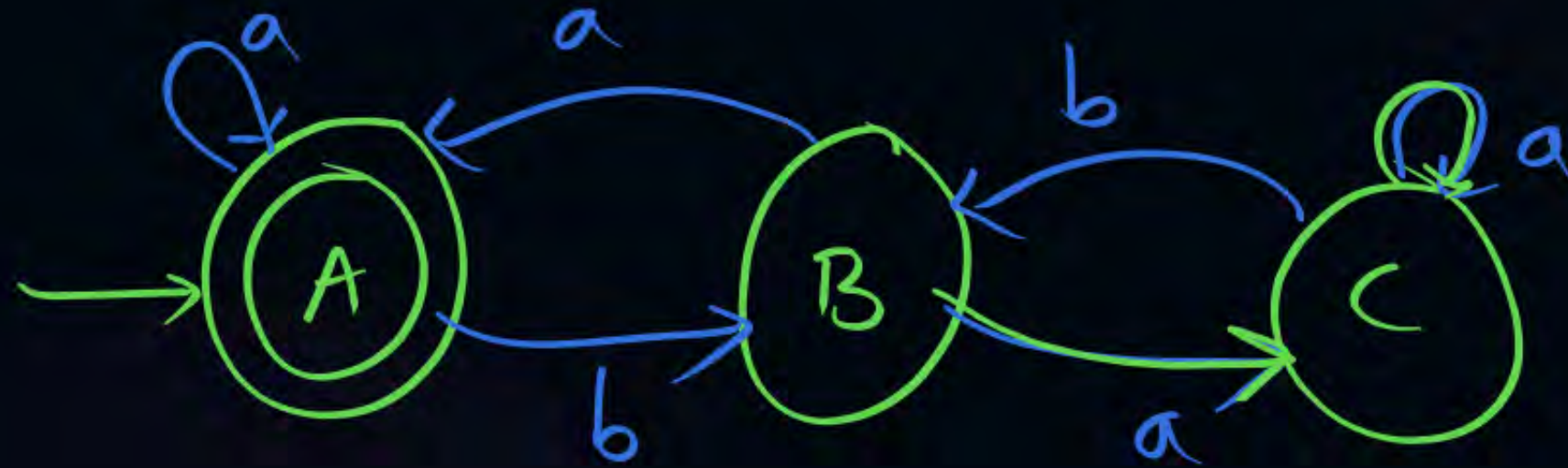
#Q. Construct Regular Expression for the following Finite Automata.



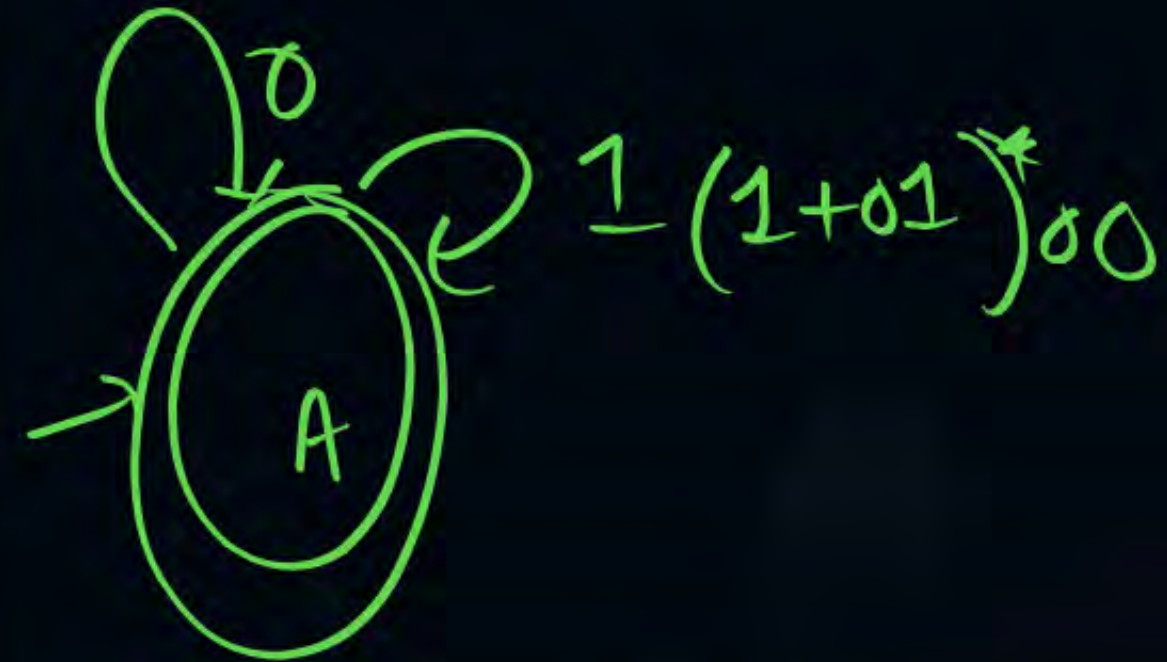
#Q. Construct Regular Expression for the following Finite Automata.



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#Q. Construct Regular Expression for the following Finite Automata.



$$\underline{\underline{[0 + 1(1+01)^*00]^*}}$$



THANK - YOU